

VALLADOLID, Spain

WMO: 081410

Lat: 41.6408N

Lon: 4.7544W

Elev: 2411

StdP: 13.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	27.6	18.6	15.3	33.3	21.7	17.8	34.7	21.7	44.5	18.4	43.4	3.1	20	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.4	94.2	64.7	91.3	64.1	88.1	63.2	67.0	87.7	65.8	86.0	64.6	83.9	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.7	86.8	70.9	59.3	82.3	69.9	58.0	78.6	69.0	32.8	88.0	31.8	86.2	30.8	83.8	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.7	11.6	21.3	99.0	2.6	2.1	19.5	100.5	18.0	101.8	16.5	103.0	14.6	104.5		
(5)			20.4	70.7	2.6	3.2	18.5	73.0	17.0	74.9	15.6	76.7	13.7	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	40.3	42.6	48.1	52.1	58.9	67.8	72.4	71.8	65.0	56.9	46.3	40.8	(6)
(7)		DBStd	12.86	5.45	5.34	5.72	6.18	6.69	6.93	5.38	5.34	5.76	5.84	5.85	5.73	(7)
(8)		HDD50	1114	302	210	104	48	8	0	0	0	0	12	141	289	(8)
(9)		HDD65	4218	765	626	525	389	209	51	8	8	69	256	562	750	(9)
(10)		CDD50	3056	2	4	45	110	285	533	693	676	450	226	28	4	(10)
(11)		CDD65	687	0	0	0	1	22	134	237	219	68	6	0	0	(11)
(12)		CDH74	8447	0	0	1	40	353	1691	2912	2589	785	76	0	0	(12)
(13)		CDH80	3579	0	0	0	1	87	722	1368	1165	228	8	0	0	(13)
(14)	Wind	WSAvg	4.7	4.5	5.0	5.4	5.3	4.9	5.0	4.8	4.6	4.3	3.9	4.5	4.0	(14)
(15)	Precipitation	PrecAvg	17.5	1.8	1.3	1.3	1.7	1.9	1.3	0.6	0.6	1.2	2.0	2.0	1.7	(15)
(16)		PrecMax	27.6	7.2	4.1	5.1	4.6	6.4	3.7	2.6	2.7	4.2	6.3	6.5	6.2	(16)
(17)		PrecMin	7.5	0.0	0.0	0.0	0.1	0.3	0.0	0.0	0.0	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	4.4	1.5	1.0	1.2	1.1	1.2	0.9	0.7	0.6	1.1	1.4	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	63.0	71.1	78.3	87.3	97.1	97.5	97.1	90.9	80.6	65.7	56.9	(19)
(20)			MCWB	50.5	49.8	52.5	57.1	62.6	65.4	65.3	65.6	63.3	58.7	56.6	52.0	(20)
(21)		2%	DB	52.7	58.2	67.2	74.2	82.2	92.2	94.1	93.6	86.0	75.7	60.8	53.9	(21)
(22)			MCWB	48.6	48.4	51.1	55.7	61.1	64.0	64.8	64.5	61.9	58.5	53.6	50.2	(22)
(23)		5%	DB	50.7	54.9	63.8	69.6	77.9	87.8	91.2	90.5	82.1	71.2	57.6	51.5	(23)
(24)			MCWB	47.5	47.1	50.0	54.6	59.4	62.9	64.3	63.8	60.8	57.1	52.2	48.5	(24)
(25)		10%	DB	48.5	52.0	60.0	65.0	73.5	83.9	87.9	86.8	78.5	67.5	55.1	49.4	(25)
(26)			MCWB	45.6	45.8	49.0	52.4	57.4	62.1	63.3	63.0	59.9	56.3	50.7	46.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	52.7	55.2	60.1	65.3	67.8	69.7	68.8	65.6	62.8	57.9	53.5	(27)
(28)			MCDWB	54.0	58.6	65.8	74.1	82.8	87.1	90.4	88.7	83.2	72.5	62.7	55.1	(28)
(29)		2%	WB	49.9	50.3	53.0	57.2	62.3	66.0	67.2	66.7	63.9	60.6	55.2	51.0	(29)
(30)			MCDWB	51.9	55.8	62.4	70.5	78.5	86.8	88.0	87.8	81.1	69.9	59.2	53.2	(30)
(31)		5%	WB	47.9	48.3	51.4	55.0	60.3	64.5	65.8	65.4	62.5	59.1	53.2	49.0	(31)
(32)			MCDWB	50.1	53.3	60.9	67.0	75.2	83.9	86.3	85.7	78.0	67.4	56.6	51.2	(32)
(33)		10%	WB	46.0	46.6	49.8	53.1	58.4	63.0	64.4	64.1	61.0	57.6	51.3	47.0	(33)
(34)			MCDWB	48.2	51.4	58.4	63.7	71.1	80.8	83.8	83.2	75.2	65.2	54.5	49.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	16.4	19.7	20.2	21.9	25.3	27.4	26.8	23.4	18.6	13.5	11.8	(35)
(36)			MCDBR	14.0	21.7	27.5	28.7	29.4	31.4	32.0	31.5	29.4	25.6	17.2	12.6	(36)
(37)			MCWBR	10.0	12.9	14.1	13.4	11.5	10.0	9.8	10.2	10.7	11.4	10.7	9.2	(37)
(38)		5% WB	MCDBR	11.4	18.3	22.6	25.3	26.4	28.6	29.4	28.4	25.0	19.6	14.4	11.0	(38)
(39)			MCWBR	8.9	11.9	12.4	12.4	11.2	10.2	10.1	10.3	9.8	9.7	9.9	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.294	0.324	0.336	0.347	0.352	0.337	0.342	0.330	0.318	0.294	0.280	(40)	
(41)		taud	2.519	2.485	2.417	2.392	2.385	2.385	2.400	2.382	2.434	2.491	2.526	2.528	(41)	
(42)		Ebn at Noon	279	293	294	296	294	292	295	291	288	279	272	271	(42)	
(43)		Edh at Noon	23	27	33	36	37	37	36	36	32	27	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	569	897	1292	1663	2026	2295	2391	2088	1606	1046	660	516	(44)	
(45)		RadStd	63	100	151	170	176	96	92	91	77	99	80	51	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) -0.72	(h) -0.86	(i) N/A	(j) N/A	(k) N/A	(l) +110
(47) Regional (5 neighbors)		N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	+99

Nomenclature: See separate page